Krishnaa **Vadivel** | Computational Physics | Distributed Computing | Machine Learning

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Profile

Computational physics PhD researcher with training across condensed matter, numerical methods, distributed scientific software, and machine learning for accelerated calculation. Applies those tools in first-principles studies of self-trapped excitons and excitonic polarons, many-body theory for electron-phonon coupled systems, and scalable implementations on national-lab HPC platforms.

Research Experience

Yale University, Electrical Engineering
Research Assistant / PhD Researcher *2021–present*

- Developed distributed scientific software in Python and C++ with MPI, OpenMP, PETSc, and SLEPc for large linear-algebra and simulation workloads as part of an INCITE award using exascale resources on national-lab HPC systems.
- Developed many-body and field-theoretic formulations of coupled electron, exciton, and phonon systems, including coherent-state and path-integral approaches for time-dependent and self-localized states.
- Ported and optimized computational workloads for accelerator-enabled HPC environments using CUDA and ROCm, with experience running on Frontier at OLCF, Perlmutter at LBNL, and other leadership-class clusters.
- Used PyTorch, PyTorch Geometric, and equivariant graph neural networks to accelerate self-trapped-exciton calculations and related materials simulations.
- Packaged scientific applications in Docker containers for deployment across different computing environments and used TotalView, Linaro DDT, Bash, and Linux command-line tools to debug distributed applications.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026 on self-trapped exciton formation and related excited-state materials physics.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer *2019–2021*

- Built CUDA-based data-processing pipelines and custom GPU-enabled engineering tooling for large acoustic waveform datasets generated by downhole logging tools.
- Built full-stack server applications with C#, ASP.NET, and Microsoft SQL Server for database design, real-time remote tool testing, and calibration of temperature and pressure sensors.
- Applied finite-element analysis with dolfinx for frequency analysis of steel pipes in well environments to support in-house physics and engineering studies.

FindLight
Photonics Technical Writer *2018–2019*

- Produced technical articles and product-facing photonics content covering lasers, optics, and optoelectronic components for a specialized engineering readership.

Selected Projects

Distributed First-Principles Exciton-Phonon Simulation

- Developed distributed first-principles calculations for excited-state and exciton-phonon problems, with an emphasis on self-trapped excitons, many-body theory, and scalable materials simulation.

Equivariant Graph Neural Networks for Molecular Systems

- Developed PyTorch Geometric equivariant models for molecular-property prediction and explored their use for accelerating scientific calculations.

Agentic Scientific Input Generation

- Built retrieval-driven tooling that reads software documentation and research papers to generate structured computational-science inputs and starter configurations for physics and chemistry codes.

Integrated Photonics and FDTD Modeling

- Designed photonic structures including nanophotonic ring-resonator band-pass filters in telecom bands with Meep-based FDTD simulation.

dolfinx Finite-Element Analysis

- Applied dolfinx-based finite-element analysis to frequency studies of steel pipes in well environments.

Education

Yale University
PhD, Electrical Engineering *2021–present*
Ongoing doctoral research in computational materials, many-body theory, and scientific computing.

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*
Included work in stochastic computing, device-oriented EE topics, and advanced design coursework.

Cornell University
BS, Electrical and Computer Engineering *2013–2017*
Included embedded systems, robotics, and core mathematics, physics, and computing foundations.

Selected Publications

2025: de Paula, A. M., Li, S., Hou, B., Vadivel, S., Teles-Ferreira, D. C., Iudica, A., Kabacinski, P., Hosseini, H., McArthur, J., Cerullo, G., et al. (2025). Time-domain observation of ultrafast self-trapped exciton formation in lead-free double halide perovskites. *Journal of the American Chemical Society, 147*(32), 28923–28931.

Selected Conference Presentations

2026: Vadivel, S., Grande, R. D., Strubbe, D. A., & Qiu, D. Y. (2026). First-principles study of exciton-polarons and self-trapped excitons. *APS Global Physics Summit 2026*.

2025: Vadivel, S., Qiu, D. Y., Grande, R. D., & Strubbe, D. A. (2025). First-principles study of self-trapped exciton formation in double perovskites. *APS Global Physics Summit 2025*.

2024: Vadivel, S., & Qiu, D. (2024). First-principles study of self-trapped exciton formation in double perovskites using excited state forces. *APS March Meeting Abstracts 2024*, F01–007.

Selected Coursework

Mathematics: Differential Geometry; Abstract Algebra; Honors Mathematical Analysis II

Computer Science: Theoretical Computer Science; Probabilistic Machine Learning

Physics: Graduate Quantum Mechanics I–II; Solid State Physics I–II; Many-Body Quantum Physics

Electrical Engineering: Semiconductor Physics; Thin Film Science; Lasers and Optoelectronics; Integrated Photonics; VLSI Design; Analog VLSI Design; Mathematics of Signals and Systems; Advanced Embedded Systems

Technical Skills

Languages: Python, C, C++, CUDA, JavaScript, Bash, PowerShell, SQL, C#

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, pandas, SciPy, SymPy, matplotlib, mpi4py, petsc4py, slepc4py, dolfinx

Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks

Systems: Linux command line, macOS, Windows, CMake, Docker, Docker Compose, Kubernetes, gcloud, ASP.NET, Microsoft SQL Server

Debugging / Platforms: TotalView, Linaro DDT, distributed-application debugging, Frontier, Frontera, Perlmutter, and related national-lab or university HPC environments